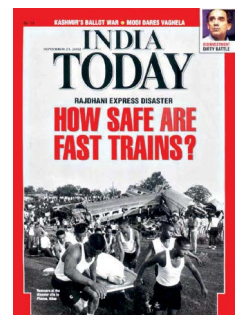
FROM THE

EDITOR-IN-CHIEF

After a catastrophe like the triple train collision that occurred near Balasore in Odisha on June 2, thoughts inevitably turn to rail safety. To understand the nature of the challenge, one must grasp the sheer enormity of the Indian Railways network. If you lay all of India's running railway tracks in a single line around the Equator, you can go around the earth twice. A whopping 102,831 km, even if you exclude auxiliary bits like sidings and yards. Include those and you can finish the third trip around the planet with a couple of thousand miles to spare. Last year, it carted nearly *four times the Indian population* across overpopulated cities, booming towns and small kasbahs strewn along 68,043 km of routes crisscrossing the vastness of a still-largely rural subcontinent. In 2018-19, the pre-pandemic peak, the total number of passengers nearly touched *eight billion*. Stack on to that some 1,400 million tonnes of freight, moving between buzzing economic centres and bases of natural resources to points of despatch. The Indian Railways is a behemoth of an organisation with over 1.1 million employees and is the largest public sector employer. It is a ministry of the Union government, which means it has to deal with political pressures on various fronts—from subsidised fares, mollycoddling favourite sectors and recommendations for jobs. Yet, it has to run with the efficiency of a private corporation because of its criticality for the economy while maintaining public safety. What's called for, therefore, is a nearly superhuman effort to bring down to zero the statistical probability of a mishap across billions of points of vulnerability.



September 23, 2002

The scale of the horror is unmappable when something goes wrong. The basic details of the Balasore tragedy are well-known by now. The Shalimar-Chennai Coromandel Superfast—hurtling along at 128 kmph near Bahanaga Bazar, a small station 180 km north of the capital Bhubaneswar—exited the main track and entered a loop line. As the name implies, this is a small side track, typically running about 800 metres, on which slower trains are parked momentarily while priority trains get right of way on the main track. The southbound express train, with its usual microcosm of Indian society on board, entered such a line and crashed headlong into a stationary, iron ore-laden goods train. Eyewitnesses spoke of bogies almost flying in the air for a few seconds and the tail end of a third train coming up the opposite direction, the Yeshwantpur-Howrah Superfast, got caught in that whirlpool of tonnes of twisted, flying metal.

It was one of the worst disasters in Indian railway history. Locals resorted to extreme metaphors to convey what they saw, felt and heard. It was like an earthquake, one of them told INDIA TODAY. Another likened it to a nuclear bomb. A third thought for a moment that a stray missile from the nearby DRDO test range Chandipur-on-Sea had hit sleepy Bahanaga. The grim statistics—288 dead, more than a thousand injured—barely convey the cumulative trauma. After the initial burst of speculation, with its usual mix of conspiracy theories, Union rail minister Ashwini Vaishnaw said there was enough clarity on the mechanical and human processes involved. Expect more forensic detailing soon from the CBI, which is now investigating.

The INDIA TODAY team pooled its research and domain knowledge, talking to everyone from the seniormost figures in the ministry to local eyewitnesses to piece together an exhaustive and exclusive narrative of what happened for this week's cover story. What caused the mishap? Which cog in the wheel came loose? What lessons can we learn from the Coromandel crash to ensure India never has to witness one more tragedy of this kind?

The fatal mistake occurred at the last node of the electronic interlocking system, where it interfaces with the human actors. This system controls the inter-track connectors and, thus, decides which track trains move on. A boom barrier at a level crossing near Bahanaga was malfunctioning and had to be replaced. By standard protocol, this warrants a temporary cessation of movement on the tracks. Work to instal a new one was going on for a week. This being one of India's busiest sectors, working at nearly 100 per cent capacity, they opted for the 'disconnect' mode at the location box—the last point in the electronic system. This delinks the boom barrier signals from the interlocking system and keeps the tracks running. But as the system returned to normal, the signal inspector on the ground plugged in the wires wrong, switching on the loop circuit instead of the main track. Tragically, he and the station master missed doing the crucial physical verification to see if everything was in sync after the new boom barrier was installed. Four minutes later, at 6.55 pm, the Coromandel Express sped uncontrollably into the loop line to its hellish end.

The scenario raises a number of questions and offers sobering lessons—both specific and general. After a decade of being put into service, the electronic interlocking system now covers about half of India's 7,325 stations. Even if it did not fail by itself, to be deemed totally fail-safe, it has to be immunised against single-point error or manipulation. Every act must necessarily pass through multiple filters, with the system itself jamming and alerting human monitors. Overall, India has seen a sharp decline in major rail mishaps. "Accidents per million train kilometres has come down from 0.65 in 2001 to 0.03 in 2022," says a former Member (Traffic) of the Railway Board. But India still logs among the highest tolls in the world in rail accidents.

A Comptroller and Auditor General (CAG) report in December 2022 red-flagged severe staffing shortage as a contributory factor—many of the 312,000 non-gazetted vacancies minister Vaishnaw admits to are in the safety category. A vital crew like loco pilots, motormen, guards and others involved in station, track and rolling stock operations end up doing dangerously long shifts lasting beyond 12 hours. Track inspections fall short by up to 100 per cent in some sectors. With automation coming in and the government eager to downsize, the railways are caught in a vulnerable transition period, and many jobs are parcelled out to ill-trained contractual staff. The government is focused on creating a better service at various levels, bringing in faster, swankier trains, upgrading stations, smoothening the passage of freight, and installing a 5G-driven anti-collision system like Kavach et al. These are worthy objectives, but the Railways has to manage the urgently-needed modernising process without endangering passenger safety.

(Aroon Purie)